

Math 6601 Homework

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Problem 1.

Proof. By the Weierstrass approximation theorem, there is a sequence of polynomials $\{P_n\}$ that converges to f uniformly on $[0, 1]$. Let $\varepsilon > 0$ be given. Then we can choose m such that $|P_m(x) - f(x)| \leq \frac{\varepsilon}{2}$ for all $x \in [0, 1]$. Hence, if N is the degree of P_m , then for any $n \geq N$ we must have

$$\begin{aligned} |E_n(f)| &= \left| \int_0^1 f(x) \, dx - \sum_{i=1}^n w_i f(x_i) \right| \\ &\leq \int_0^1 |P_m(x) - f(x)| \, dx + \sum_{i=1}^n w_i |f(x_i) - P_m(x_i)| + \left| \int_0^1 P_m(x) \, dx - \sum_{i=1}^n w_i P_m(x_i) \right| \\ &\leq \int_0^1 \frac{\varepsilon}{2} \, dx + \sum_{i=1}^n \frac{\varepsilon}{2} w_i + |E_n(P_m)| \\ &\leq \varepsilon \end{aligned}$$

because, by assumption, $E_n(P_m) = 0$, and

$$0 = E_n(1) = \int_0^1 1 \, dx - \sum_{i=1}^n w_i \cdot 1 = 1 - \sum_{i=1}^n w_i,$$

which implies that $\sum_{i=1}^n w_i = 1$. □

Problem 2.

Proof. We begin by observing that, by the product rule,

$$\psi'(x) = \sum_{k=0}^n \prod_{\substack{i=0 \\ i \neq k}}^n (x - x_i).$$

Hence,

$$\psi'(x_k) = \prod_{\substack{i=0 \\ i \neq k}}^n (x_k - x_i) \neq 0.$$

It follows that, if $x \neq x_k$, then

$$L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i} = \frac{1}{\psi'(x_k)} \prod_{\substack{i=0 \\ i \neq k}}^n (x - x_i) = \frac{\psi(x)}{(x - x_k)\psi'(x_k)}.$$

Thus, for $x \neq x_k$,

$$p(x) = \sum_{k=0}^n L_k(x) f(x_k) = \psi(x) \sum_{k=0}^n \frac{f(x_k)}{(x - x_k) \psi'(x_k)}.$$

□

Problem 3.

Let $z_i = e^{x_i}$ for $i = 0, 1, \dots, n$. Then $z_0, z_1, \dots, z_n$ are unique because $x_0, x_1, \dots, x_n$ are unique and the exponential function is injective; therefore, there is a unique polynomial $Q_n(z) = \sum_{k=0}^n b_k z^k$ such that $Q(z_i) = y_i$ for $i = 0, 1, \dots, n$. Choose $c_i = b_i$ for $i = 0, 1, \dots, n$. Then

$$P_n(x) = \sum_{k=0}^n c_k e^{kx} = \sum_{k=0}^n b_k (e^x)^k = Q_n(e^x),$$

so $P_n(x_i) = Q_n(e^{x_i}) = y_i$. If $p_n(x) = \sum_{k=0}^n C_k e^{kx}$ satisfies $p_n(x_i) = y_i$ for $i = 0, 1, \dots, n$, then $q_n(z) = \sum_{k=0}^n C_k z^k$ satisfies

$$q_n(z_i) = \sum_{k=0}^n C_k z_i^k = \sum_{k=0}^n C_k e^{kx_i} = p_n(x_i) = y_i$$

for $i = 0, 1, \dots, n$. Hence, q_n and Q_n are the same polynomial by the uniqueness of Q_n . Then $c_i = C_i$ for $i = 0, 1, \dots, n$, which implies that $P_n = p_n$. Thus, P_n is unique.

Problem 4.

- (a) Let $f(x) = 1$. Then f is a polynomial of degree 0, so it is its own unique interpolating polynomial P_n on $x_0, x_1, \dots, x_n$ of degree at most n for any $n \geq 0$. Since P_n is also given in terms of Lagrange polynomials, we have

$$1 = f(x) = P_n(x) = \sum_{k=0}^n f(x_k) L_k(x) = \sum_{k=0}^n L_k(x).$$

- (b) Assume that $n \geq 1$. Recalling that $W(x) = \prod_{i=0}^n (x - x_i)$, given $x \in [0, 1]$, it follows that there is some $i \in \{0, 1, \dots, n-1\}$ such that $|x - x_i| \leq \frac{1}{n}$ and $|x - x_{i+1}| \leq \frac{1}{n}$. On the other hand, $|x - x_j| \leq 1$ for all $j = 0, 1, \dots, n$. Hence,

$$|W(x)| \leq \frac{1}{n^2}, \quad x \in [0, 1].$$

This implies that

$$|f(x) - P_n(x)| = \left| \frac{f^{(n+1)}(\xi(x)) W(x)}{(n+1)!} \right| \leq \frac{K(n+2)!}{(n+1)! n^2} \leq K \frac{n+2}{n^2}, \quad x \in [0, 1].$$

Thus, $\|f - P_n\|_\infty \leq K \frac{n+2}{n^2}$. Since $K \frac{n+2}{n^2} \rightarrow 0$ as $n \rightarrow \infty$, given $\varepsilon > 0$ we can choose N such that $n \geq N$ implies that $\|f - P_n\|_\infty < \varepsilon$.